DATASCI 530 – Assignment 5

For this assignment, you will practice using some of the machine learning methods we have used in class so far. We will be working with the Iris dataset, which contains information about different flower characteristics. You can download the data here.

https://archive.ics.uci.edu/dataset/53/iris

Your goal is to predict the "class" label.

1. Start off with a description of the dataset. How many rows are there? How many features? What is the proportion of observations in each class?
2. Run a comparison of different machine learning methods, which should include support vector machines (SVMs), random forests, neural networks, and at least one method we have not covered in class (lasso, naïve bayes, ridge, etc.)
    - Evaluate the predictive performance using cross-validation. Consider more than one evaluation metric.
    - Use the default tuning parameters for all methods except one (you choose). For this method, define a grid of tuning parameters and select the best-performing ones.

New insight question:

3. Compute an additional table/figure not included in the questions above.
4. What do we learn from it?

Make sure to properly document your approach. Give a recommendation about which method you think predicts the class labels best.

Deliverable: Submit an HTML file with your results on Canvas. Include Markdown text chunks to write comments on your findings and explain your methodology. You are expected to have a title for your project and a short section at the beginning summarizing the findings. We recommend that you do not print raw data and only print final tables and/or figures.

When submitting your work, imagine that you are sharing your report with colleagues who have asked you to explain and justify the key decisions and findings from your analysis. The file should be professionally presented, self-explanatory, and have clearly labeled sections.

Rubric:

You will be graded both on the code and the writing using the following point system:

Code - Code runs without issues. - Code is appropriate for answering the questions. - File is appropriately organized, with different sections and comments clarifying the definition of variables and dataset names.	2 points
Description of methodology - Key coding decisions are explained via Markdown chunks. - Choices about how to clean the data are explained. - Quality checks are conducted, such as verifying the sample size of merged datasets, checking for the presence/absence of missing values for key variables, and/or computing any relevant descriptive statistics.	2 points
Results and Visualization - Includes a short summary of the findings. - Results are correctly interpreted and easy to follow. - Includes either tables or graphs to display results from different groups. - All graphs look professional (have axis and title labels, a suitable color scheme, appropriate font sizes, a legend when needed, etc.). - Tables are properly formatted (e.g., have appropriately rounded values, relabeled columns, captions, and are not too long).	4 points
New Insight - Analysis is distinct from the questions already provided. - Clear interpretation of the new analysis.	2 points